Cloud Infrastructure Assessment Workshop

Troubleshooting and Review Session

- Prepare for assessment success by building practical skills in cloud ML setups.
- Focus on Azure VM configuration, networking, scaling, and Bash scripting.

Learning Objectives

- Review key concepts in cloud GPU VM configuration and automation.
- Practice Bash scripting for managing environments and jobs on Azure.
- Troubleshoot common setup errors and resolve infrastructure issues.
- Document assessment evidence and polish your portfolio.

Workshop Roadmap

1. Hands-on review; Azure GPU VM and network setup.
2. Practice; Bash scripting for job and environment management.
3. Troubleshooting; Resolve common assessment blockers.
4. Performance review; What counts as strong evidence.
5. Peer support; Collaborative debugging.
6. Submission checklist and Q and A.

Recap; Previous Learnings

- Bash and shell scripting basics; automate ML workflows. (Weeks 2-4)
- Azure VMs for machine learning; setup and access. (Week 7)
- Cloud networking and security; scaling and automation. (Weeks 8, 12)
- Running PyTorch on GPU; troubleshooting and optimisation. (Week 13)

Review; Configuring Azure GPU VMs

- Select the right GPU VM for model training workloads.
- Use the Azure portal or CLI for provisioning.
- Set up SSH keys and role permissions for secure access.
- Attach storage for datasets and model checkpoints.
- Ensure GPU drivers and PyTorch are installed and verified.

Practical Example; Azure CLI VM Deployment

- Create a resource group with `az group create`.
- Deploy a VM using `az vm create` with GPU-enabled size.
- Open necessary ports for SSH and Jupyter access.
- Verify deployment; connect via SSH and check `nvidia-smi`.

Networking and Autoscaling; What to Watch For

- Virtual Networks must allow internal traffic for distributed jobs.
- Network Security Groups must permit essential connections only.
- Use Azure Scale Sets for autoscaling larger workloads.
- Monitor resource usage to avoid unexpected runtime errors.

Bash Scripting; Core Patterns

- Automate VM management using scripts; start, stop, status.
- Use loops to submit and monitor multiple ML jobs.
- Handle environment variables for flexible script setups.
- Redirect logs with `>>` to keep track of job outputs and errors.

Troubleshooting; Common Mistakes

- Access issues; check SSH keys and role permissions.
- Software errors; ensure CUDA drivers and Python environments match.
- Resource errors; confirm correct VM and scale set selections.
- Firewall blocks; review port settings in network security groups.
- Bash script errors; use `echo`, `set -x`, and log inspection.

Activity; Test and Document Your Configuration

- Carry out a complete Azure GPU VM setup using the CLI or portal.
- Deploy a test workload and capture outputs; e.g. `nvidia-smi`, Python version, disk space.
- Take screenshots of major steps and outcomes.
- Save scripts and terminal logs for your assessment portfolio.

Assessment Evidence; What to Include

- Screenshots showing successful VM and network provisioning.
- Example Bash scripts with comments and test logs.
- Performance outputs; GPU and memory status, task runtimes.
- Short notes; troubleshooting steps and how issues were resolved.

Peer Support; Collaborative Debugging

- Share blockers and workarounds with class peers.
- Test files on each other's VMs to check reproducibility.
- Discuss how you solved unique cases; permissions, network errors, script bugs.

Review; Assessment Checklist

- All required components are complete; VMs, networks, scripts, logs, and screenshots.
- Scripts run without errors and deliver expected outputs.
- All troubleshooting steps are documented.
- Portfolio is organised and ready for submission.

Next Steps

- Finalise documentation and double-check assessment criteria.
- Continue practicing with scale sets, automation, and performance checks.
- Prepare any outstanding questions for the lecturer.
- Weekly goal; ready for single-GPU PyTorch project next week.

Questions and Wrap-Up

- Any blockers with Azure, Bash, or networks?
- Need extra guidance preparing your evidence?
- Next week; move from infrastructure to advanced GPU workflows.
- Reach out for support; in class or by appointment.